

Type	Name	Value	Units	Rule	Formula	Expose
Dim	R_Thread	47.50	mm	Formula	= R_SmallEnd - Spline_Depth - Thread_TaperSpline_Clearenc	☑
Dim	R_SmallEnd	53.00	mm			☑
Dim	L_Taper	300.00	mm			☑
Dim	T_Flange	45.00	mm			☑
Dim	L_Hollow	1155.00	mm	Formula	= L_Total - T_Flange - L_Solid_Thread_Taper_Length	☑
Dim	IR_Shaft	30.00	mm			☑
Dim	OR_Shaft	62.50	mm			☑
Dim	OR_Flange	137.50	mm	Formula	= OR_Shaft *2.2	☑
Dim	L_Solid_Thread_Taper_Length	400.00	mm	Formula	= L_Thread + L_Taper	☑
Dim	Spline_Half_Angle_Smallend	15.00	°	Formula	= Spline_Angle_Smallend /2	☑
Dim	Spline_Angle_Smallend	30.00	°	Formula	=360/(No_Spline + No_Space_Between_Spline)	☑
Dim	OD_Spline_Smallend	106.00	mm	Formula	= R_SmallEnd *2	☑
Dim	ID_Spline_Smallend	96.00	mm	Formula	=(R_SmallEnd - Spline_Depth)*2	☑
Dim	Width_Bearing1	130.00	mm	Formula	= Width_Bearing	☑
Dim	Width_Half_Bearing1	65.00	mm	Formula	= Width_Bearing /2	☑
Dim	Width_Bearing2	130.00	mm	Formula	= Width_Bearing	☑
Dim	Width_Half_Bearing2	65.00	mm	Formula	= Width_Bearing /2	☑
Dim	L_Bearing2_From_Bearing1	755.00	mm	Formula	= L_Hollow - L_Bearing1_From_Spline *2	☑
Dim	L_Bearing1_From_Spline	200.00	mm			☑
Dim	Spline_Half_Angle_Biggerend	15.00	°	Formula	= Spline_Half_Angle_Smallend	☑
Dim	Spline_Angle_Biggerend	30.00	°	Formula	= Spline_Angle_Smallend	☑
Dim	ID_Spline__Biggerend	115.00	mm	Formula	=(OR_Shaft - Spline_Depth)*2	☑
Dim	OD_Spline_Biggerend	125.00	mm	Formula	= OR_Shaft *2	☑
Dim	Shaft_Hole_Fillet_Radius	5.00	mm			☑
Dim	Spline_To_Shaft_Fillet_Radius	5.00	mm			☑
Dim	Spline_Cut_Smallend_Sketch_Line1	72.50	mm	Formula	= OR_Shaft +10	☑
Dim	Spline_Cut_Smallend_Sketch_Line2	72.50	mm	Formula	= OR_Shaft +10	☑
Dim	Spline_Cut_Biggerend_Sketch_Line1	72.50	mm	Formula	= OR_Shaft +10	☑
Dim	Spline_Cut_Biggerend_Sketch_Line2	72.50	mm	Formula	= OR_Shaft +10	☑
Dim	L_Thread	100.00	mm			☑
Dim	Bearing_Surface_Seperating_Planes_FiniteDept	135.00	mm	Formula	= OR_Shaft *2+10	☑
Dim	Flange_PCD	200.00	mm	Formula	=(1.6* OR_Shaft)*2	☑
Dim	L_Bearing2_Edge2_Plane	135.00	mm	Formula	= OR_Shaft *2+10	☑
Dim	L_Bearing2_Edge1_Plane	135.00	mm	Formula	= OR_Shaft *2+10	☑
Dim	L_Bearing1_Edge2_Plane	135.00	mm	Formula	= OR_Shaft *2+10	☑
Dim	L_Bearing1_Edge1_Plane	135.00	mm	Formula	= OR_Shaft *2+10	☑

Exposed Name	Comment
Thread Radius	
Taper SmallEnd Radius	
Taper Length	
Flange Thickness	
Hollow Length	
Shaft Inner Radius	
Shaft Outer Radius	
Flange Outer Radius	
Solid thread + Taper Section Length	
Spline Half Angle Smallend	
Spline Angle Smallend	
Spline Outer Diameter Smallend	
Spline Inner Diameter Smallend	
Bearing1 Width	
Bearing1 Half Width	
Bearing2 Width	
Bearing2 Half Width	
Length Of Bearing2 From Bearing1	
Length Of Bearing1 From Spline	
Spline Half Angle Biggerend	
Spline Angle Biggerend	
Spline Inner Diameter Biggerend	
Spline Outer Diameter Biggerend	
Shaft Hole Fillet Radius	
Spline To Shaft Fillet Radius	
Spline Cut Smallend Sketch Line1	
Spline Cut Smallend Sketch Line2	
Spline Cut Biggerend Sketch Line1	
Spline Cut Biggerend Sketch Line2	
Thread Length	
Bearing Surface Seperating Planes From Shaft	
Pitch Circle Diameter	
Length Of Bearing2 Edge2 Plane	
Length Of Bearing2 Edge1 Plane	
Length Of Bearing1 Edge2 Plane	
Length Of Bearing1 Edge1 Plane	

Type	Name	Value	Units	Rule	Formula	Expose
Dim	Thread_To_Spline_Fillet_Radius	3.00	mm			☑
Dim	Shaft_To_Splange_Fillet_Radius	10.00	mm			☑
Dim	Flange_Bolt_Hole_Dia	20.00	mm			☑
Var	PhysicalProperties_Density	0.000	kg/m^3	Limit		☑
Var	PhysicalProperties_Accuracy	0.990		Limit		☑
Var	L_Total	1600.00	mm			☑
Var	Spline_Depth	5.00	mm			☑
Var	No_Space_Between_Spline	6.000		Formula	= No_Spline	☑
Var	Shaft_Taper_Angle	1.81	°	Formula	=atn((OR_Shaft - R_SmallEnd)/ L_Taper)*(180/3.14159)	☑
Var	Width_Bearing	130.00	mm	Formula	= OR_Shaft *2+ Width_Bearing_Tolerance	☑
Var	Width_Bearing_Tolerance	5.00	mm			☑
Var	No_Spline	6.000		Limit		☑
Var	Thread_TaperSpline_Clearence	0.50	mm			☑
Var	No_Flange_Bolts	6.000		Limit		☑

Exposed Name	Comment
Thread To Spline Fillet Radius	
Shaft To Splange Fillet Radius	
Bolt Holes Diameter	
Density	
Accuracy	
Shaft Total Length	
Spline Depth	
Number of Spaces Between Spline	
Shaft Taper Angle	
Bearing Width	
Bearing Width Tolerance	
Number Of Spline	
Thread Taper And Spline Clearence	
Number Of Bolts In Flange	